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Technology-mediated learning paradigm and the blended learning ecosystem: what works for working professionals?

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Abstract

While the need to continually reskill and upskill is universally acknowledged by working professionals and organizations, the constraints of location, time and antiquated learning formats prove to be significant hurdles in any learning endeavor. In such a scenario, blended learning has emerged as a viable and increasingly popular option among working professionals. This research paper analyses the conceptual and contextual relevance of blended learning for working professionals and makes three notable contributions. Firstly, it develops a learning paradigm relevant to working professionals, whose learning needs are context-centric, skills-focused, situation-specific, peer-dependent and action-oriented. Secondly, the paper operationalizes this learning paradigm through an integrative framework, the blended learning ecosystem. And finally, the paper summarizes institutional, faculty-related, student-specific and pedagogical variables considered as key success factors for effective blended learning experience.

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1. Introduction

Globally, the rapid pace at which technology and competitive landscape are changing has meant that, not only products and business models, but also skills that are required by working professionals and organizations have a reducing shelf life and are becoming redundant faster. Paulo [1] has observed that this is reflected in skill obsolescence becoming a bigger concern, as compared to even being laid off, for the majority of workers. Kaufman [2] defined Skill Obsolescence as “the degree to which professionals lack the up-to-date knowledge or skills necessary to maintain effective performance in their current or future work roles.” As per Cedefop [3], this can be categorized primarily into obsolescence of physical or cognitive skills, and economic skills, i.e.

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those that were previously useful becoming redundant or diminishing in importance. Further, this concern is more pronounced among younger employees as they are looking ahead at careers extending well into their old age amidst a more uncertain economic and technological environment than the one faced by the older generation. Reflecting this trend, Bersin [4] refers to a recent research study that found learning and career opportunities are the biggest drivers of employees' willingness to recommend their company as a great place to work for people under age 40. Echoing this concern of working professionals, in a study conducted by Deloitte [5], 84% business leaders cited the "need for improved organizational learning" as a top priority and 44% mentioned it as urgent, in a global study on human capital trends. Longenecker et al. [6] earlier study of over 1,000 managers in global corporations found that the skills gap resulting from a rapidly changing business landscape was one of the primary causes for managerial failure to deliver on the needed organizational results.

With continuous learning, and skill acquisition and development becoming a more urgent need among working professionals, there has been a parallel development in which several technology-enabled learning options and tools have emerged. These options, for e.g. massive open online courses or MOOCs, offer flexible, convenient and often free bite-sized content delivered on electronic devices, however their effectiveness and utility in targeted skill development of working professionals may vary significantly. On the other hand, as Bersin [4] notes, traditional corporate learning management systems even though customized to organizational and business context, have "remained slow, hard to use and difficult to maintain", and may even become obstacles in the way of employee development instead of enabling it. What then could be an optimal learning solution for working professionals, who are hard-pressed for time, mostly constrained by location, and unwilling to take a career break for full-time learning, but have a recurring need to acquire and develop new skills?

In such a scenario, blended learning has emerged as a popular option among working professionals. Commonly understood as a combination of traditional classroom or face-to-face (F2F) and online methodologies, Cenejac [7] observes blended learning courses offer ease of access, interactive pedagogy, and are cost-effective and flexible. However, Kaur [8] has found that while blended learning courses have several potential benefits and high contextual significance for working professionals, they are faced with technical, organizational and instructional design challenges that can derail their adoption. On this note, Moskal et al. [9] have emphasized that successful implementation of blended learning programs requires an alignment between institutional, student and faculty goals. Further, they suggest that blended learning can be difficult to implement as it challenges the status quo of the traditional learning environment in multiple ways, while simultaneously introducing mobile, internet and cloud-based technologies. Notwithstanding these challenges and hurdles, the fact remains that blended learning offers an unprecedented opportunity for working professionals to develop their skills on-the-go, provided the design and implementation of these courses consider stakeholder priorities at their core.

2. Literature review and conceptual development

2.1. Conceptual evolution of blended learning

The emergence of blended learning as a pedagogical concept is relatively recent, starting around the turn of this century, and has been enabled by the rapid penetration of the internet and web technologies. The earliest references of the term "blended learning" and attempts to use online with traditional face-to-face medium of instruction have been documented during this period in extant research conducted by Cooney et al. [10]; Voci and Young [11]; Bonk et al. [12]; and Stewart, [13]. However, in this initial period, a concerted attempt at defining and conceptualizing blended learning was still not underway. Instead, the focus of early researchers

and practitioners of blended learning remained on viewing online learning and face-to-face learning as two distinct components, operating as adjuncts to each other. During 2002-2006, which Guzer and Caner [14] refer to as the Definition Period, research in this domain was focused on developing a definition of blended learning. Making an attempt in this direction, Osguthorpe and Graham [15] proposed that:

Blended learning combines face-to-face with distance delivery systems... but it's more than showing a page from a website on the classroom screen...those who use blended learning environments are trying to maximize the benefits of both face-to-face and online methods (p. 229).

Going a step forward, they also introduced three models of blending, based on - learning activities i.e. same students indulging in different activities online as compared to in class; students i.e. mixing different students in online and class environments; and, instructors i.e. different instructors in the online environment than in classroom. Singh [16] probed deeper into the technology aspect of blended learning and suggested that it comprised various dimensions - customized and off-the-shelf content, self-paced and collaborative learning, real-time and asynchronous, structured and unstructured, offline and online, and feedback and performance support. Further, he posited and in retrospect rightly so that, with changes in technology organizations will increasingly adopt blended learning over single delivery-mode programs. Among the most cited articles on blended learning is the one from Garrison and Kanuka [17], who stated that, “blended learning is the thoughtful integration of classroom face-to-face learning experiences with online learning experiences”, while explicating on the potential of blended learning in addressing challenges faced in higher education. Importantly, their conception of blended learning as a means to achieve satisfying learning “experiences” established a learner-centric approach in terms of policy, planning, delivery and support.

In the next phase, starting around 2007 and until now, research interest on blended learning has evolved to understanding its effectiveness and the perceptions of participants. Using the MOLTA model, Deliağaoğlu and Yıldırım [18] compared the effectiveness of blended learning with traditional learning on parameters like knowledge retention, course satisfaction and attitudes, and found that even though there was no significant difference between the two, the blended environment had higher overall satisfaction. In another study, taking a quasi-experimental approach, El-Deghaidy and Nouby [19] applied the blended e-learning cooperative or BeLCA framework and found that student achievement was significantly higher in the blended group as compared to control group. Woltering et al. [20] had similar findings in their study on medical students and concluded that blended learning enhanced student motivation, satisfaction and learning outcomes. Chen and Jones [21] attempted to understand the perception of students and conducted a survey with MBA students. They found that while traditional classroom setting was preferred on clarity of instruction, blended learning led to a better appreciation of the subject and enhanced development of analytical skills. In their research with more than 1400 university students, López-Pérez et al. [22] reported positive perception for blended learning and that it led to reduced drop-out rates and raised exam pass rates. The enhanced utility of blended learning that is indicated in the earlier research is further supported by one of the most recent studies that was conducted by Marquis and Ghosh [23], and which concluded that while performance outcomes are similar for blended and traditional instruction, there is a clear student preference for the former, particularly in higher education settings.

Overall, from these studies it has emerged that students consider web-based online learning environment effective, but not a replacement for face-to-face instruction. Further, compared to traditional classroom mode, blended learning has been observed to be superior on important parameters like student satisfaction, motivation, drop-out rate, attitude, knowledge retention and development of analytical skills.

2.2. Blended vs traditional learning paradigm

The preceding literature review has summarized the evolution of empirical research on blended learning in general, which is underscored by these studies drawing a comparison between blended and traditional learning model. However, there has been no concerted effort to understand blended learning dynamics within specific learner segments. Since the contextual focus of this research paper is on working professionals, we aim to develop a learning paradigm that is suited to this learner group, and contrast it with the traditional paradigm in which learning takes place.

Hager [24] contends that the formal learning paradigm, as reflected in the traditional classroom-based instruction assumes a didactic approach, typically defined and controlled by the teacher and shows three distinctive characteristics: firstly, the individual mind is being “steadily stocked with ideas”; secondly, learning is aimed at achieving a change in beliefs; and, thirdly, it assumes transparency of learning i.e. the idea that if learning has happened, it can be made explicit. Further, this paradigm assigns ‘second-rate status’ to context-dependent, tacit or implicit, and skill-centric learning (ibid), arguably the forms of learning and knowledge acquisition that are most relevant to working professionals. Compared to this formal or traditional paradigm, we propose a new learning paradigm that is drawn from social theories of learning with extensive contributions from Engestrom [25], Lave and Wenger [26], and Wenger [27] and based on ‘learning as participation’ perspective. Lave and Wenger (ibid) contend that learning is embedded in human activity, is situation-specific, involves the ‘whole’ person (and not mind alone) and happens through active participation of learners in a community. Elaborating on this further, Wenger (ibid) notes that:

Participation here refers not just to local events of engagement in certain activities with certain people, but to a more encompassing processes of being active participants in the practices of social communities and constructing identities in relation to these communities. Participating in a playground clique or in a work team, for instance, is both a kind of action and a form of belonging. Such participation shapes not only what we do, but also who we are and how we interpret what we do (p. 6).

Delving further into the ‘learning as participation’ perspective, Engestrom (ibid) has introduced the concept of ‘expansive learning’, which is based on activity theory and has five founding principles:

1. Activity system, instead of communities of practice, is the primary unit of analysis
2. An activity system is multi-voiced, made up of multiple points of view, traditions and interests
3. Activity systems are generated and transformed over time
4. Contradictions and tensions within and between activity systems are the source of change and development
5. An expansive transformation occurs when individuals start to question and challenge existing norms within the activity system and the “object or motive of the activity are reconceptualized” (pp. 136-137) into a broader horizon of possibilities than previously

Engestrom derived the concept of expansive learning from Bateson’s theory of learning [28], which introduced three distinctive levels of learning - conditioning/acquisition, learning the hidden curriculum, and questioning/reconstruction. Expansive learning, which takes place through the third level of questioning/reconstruction, is an action-oriented and collective, rather than an individual process and is not limited to pre-defined contents and task.

The comparison between traditional and ‘learning as participation’ paradigms clearly establishes that the former is inadequate for working professionals, whose learning needs are context-centric, skills-focused, situation-specific, peer-dependent and collective rather than individual and isolated, and not strictly bound by a

pre-defined static course curriculum. Fig 1 below illustrates the difference between the two paradigms (refer to 1a and 1b), outlining the key characteristics of both. The ‘learning as participation’ paradigm builds upon the theoretical construct of social learning and expansive learning, and which we argue is well-suited for working professionals as a learner group. Further, from a contemporary perspective, it is important to consider the growing role of technology e.g. web and mobile in facilitating engagement and participation, and therefore the proposed paradigm mentions this ‘multi-modal’ nature of learning.

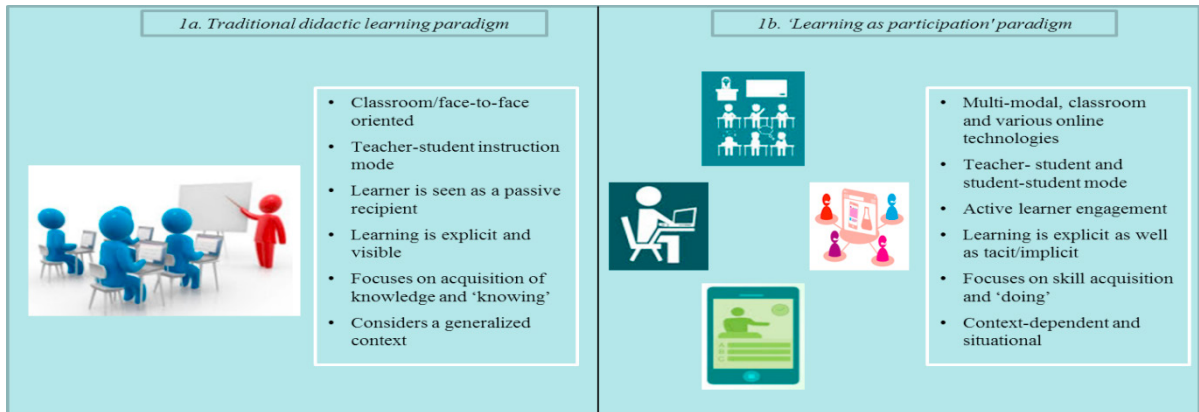


Fig 1 Comparing learning paradigms

Source: Kumar, Ranjan and Pande, Neerja (2017)

2.3. Blended learning ecosystem and key success factors

Technology-mediated learning that is highlighted in our learning paradigm (Fig 1b above) for working professionals, has become ubiquitous in professional education and vocational training. Picciano [29] observes that blended learning, which depends on digital and communication technologies to a great degree, is generally understood as the integration of face-to-face classroom with a wide variety of technology/media-based activities. However, this conceptualization of blended learning, as McGee and Reis contend [30] is “contingent upon an educational technology framework”, and does not bring centerstage what makes the blend, i.e. the current and future users of blended learning platform, both faculty and learners. Therefore, a user-centric conceptualization of blended learning has been proposed:

Blended course designs involve instructor and learners working together in mixed delivery modes, typically face-to-face and technology-mediated, to accomplish learning outcomes that are pedagogically supported through assignments, activities and assessments as appropriate for a given mode and which bridge course environments in a manner meaningful to the learner (p. 9).

In this conceptualization, there is an emphasis on the usage of mixed delivery modes, not as an end in itself but as a means to “accomplish learning outcomes”. The user-centric conceptualization and diverse learning strategy dimensions of blended learning have high significance for working professionals as a learner audience because of two reasons. One, as our earlier discussion on the learning paradigm of working professionals established, the learning needs of this learner audience are context-centric, action-oriented, participation-based and skills-focused. And two, the constraints of time, location and learning format need to be addressed for them. Both these requirements can be adequately met through a ‘blended learning ecosystem’ (refer to Fig 2), as compared to using only face-to-face or online modes. The figure illustrates how this ‘ecosystem’ provides a 360-degree learning environment for working professionals by integrating face-to-face and online modes with a

mix of human-mediated and technology-mediated learning options. Moreover, the blended learning ecosystem operationalizes the learning paradigm of working professionals (Fig 1) and its underlying theoretical constructs, social learning and expansive learning through a mix of diverse learning activities.

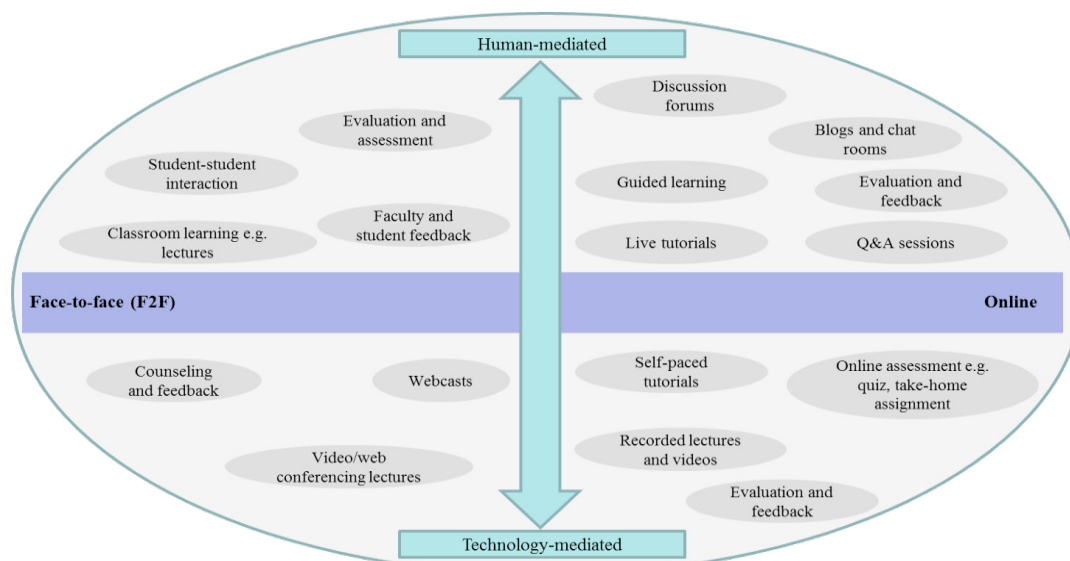


Fig 2 Blended learning ecosystem

Source: Kumar, Ranjan and Pande, Neerja (2017)

From the blended learning ecosystem illustrated above, it is evident that diverse technology and human –centric components must operate in sync for this to be an effective platform for working professionals. However, as McGee and Reis (ibid) have emphasized, blended learning is not merely about ‘blending’ or mixing delivery modes, but accomplishing learning outcomes by focusing on the learners and faculty. Dziuban and Moskal [31] have established that there is a weak statistical correlation between mode of delivery and student success or persistence. This means, effectiveness of blended learning can’t be achieved by only combining online and face-to-face delivery modes. Instead, Stacey and Gerbic [32] identify several institutional, faculty, student and pedagogic variables play an important role in ensuring successful blended learning experience. Table 1 below summarizes such key success factors along with their supporting references in extant literature and empirical research.

Table 1 Key success factors for blended learning

<i>Institutional</i>	<i>Faculty-related</i>	<i>Student-related</i>	<i>Pedagogy-related</i>
- Respond to local, community or organizational needs Sharpe <i>et al.</i>, [33] - Institutional practice of regular evaluation (ibid) - Building blocks include	- Need for continuing professional development of teachers - Vaughan, [36] - Ongoing support through membership of a community of practice - Garrison and Vaughan, [35]	- Need to develop time management and virtual learning skills of students - Vaughan, [36] - Regular and open communication on learning expectations that students are required to meet e.g.	- Pedagogical framework mapping different media to various phases of the course - Meyer, [38] - Face-to-face feedback to students on online participation, to “legitimize” the latter

<i>Institutional</i>	<i>Faculty-related</i>	<i>Student-related</i>	<i>Pedagogy-related</i>
organizational readiness, technical resources, motivated faculty, and feedback channels - Tabor, [34] • A transformative redesign process, rather than merely adding on technology - Garrison and Vaughan, [35]	• Creation of shareable and reusable digital resources to help manage teachers' workloads - Littlejohn and Pegler, [37]	online participation in discussion forums [36] • Maturity and readiness of students for independent learning [37]	as part of the course - Gerbic, [39] • Integration between the online and face-to-face environments through a sequence of activities before, during, after and in preparation for face-to-face activities [35]

3. Conclusion – implications and future direction

3.1. Key implications

The conceptual and contextual analysis conducted in this paper on the relevance of blended learning for working professionals has important implications for various stakeholders. Faculty and learning providers should consider the learning paradigm introduced in this paper, while designing and delivering blended learning programs for working professionals. More importantly, as the blended learning ecosystem establishes, those entrusted with design and delivery should not consider these programs as a mechanical 'blend' of face-to-face and online modes where the latter is added as an afterthought, but potentially an immersive and seamless learning experience with better outcome than what either of the modes could individually achieve. At the other end of the spectrum, working professionals as learners unaccustomed to the blended mode may have to first learn, how to learn in the blended learning environment. For instance, active participation in online discussion forums and various online learning activities in general, is as important as face-to-face classroom interaction. Furthermore, since blended learning relies a lot on peer-to-peer learning, working professionals have to adapt their learning styles, reset their expectations and probably even unlearn some of the habits ingrained from their previous traditional classroom-based student experience.

3.2. Future direction

Two unique contributions of this paper, the learning paradigm for working professionals and the blended learning ecosystem, can inform and guide future research in significant ways. First and foremost, the two constructs can be used to develop a conceptual model for design and delivery of blended learning programs targeted at working professionals. Secondly, the model can be used to design and conduct empirical research among working professionals attending blended learning programs, and validate hypotheses on variables that lead to effective learning outcomes, primarily skill development. Lastly, the learning paradigm and the blended learning ecosystem, which have been developed in this paper as exploratory constructs, can be enriched greatly by incorporating insights and inputs from empirical research.

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